

TARUN GUPTA

AI Systems Architect • LLM & MLOps Consultant • EU Relocation Ready

Prior Germany: Magdeburg • Hamburg • Düsseldorf | Prior EU permits: MSc work permit (2019–2022) + EU Blue Card (until Apr 2026), eligible fast-track re-issue | German A2 • English C1 • immediate availability

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PROFESSIONAL SUMMARY

- 10+ years engineering production systems; 5+ years specialised in AI/ML, GenAI, LLM, MLOps — from medical-imaging research (Universitätsklinikum Magdeburg) to applied data science (Vivere GmbH Hamburg) to 3+ years cross-industry AI/ML consulting across 5 industries (Utilities • Manufacturing • E-mobility • Retail • B2B).
- Most recent on-site: 8 months in Germany (May 2025 – Jan 2026) delivering AI/ML for German clients in Retail/E-commerce, Manufacturing, E-mobility.
- Production AI with measurable outcomes: \$100K+/month savings on LLM invoice pipeline • voice AI agents auto-resolving ~65% of inbound calls (multi-FTE workload absorbed, ~40% average handle-time reduction) • 5 forecasting pipelines from 1 codebase • 7 plug-and-play RAG chatbots.
- Cloud-agnostic MLOps: AWS (Bedrock • SageMaker • Lambda • API Gateway • Aurora • CloudWatch • EMR), Azure (OpenAI • AI Search • Document Intelligence • Blob Storage • Container Apps • DevOps), Snowflake (Cortex • Snowpark • Model Registry • Feature Store • Container Services). Certified: SnowPro Core + SnowPro Specialty Gen AI • In progress: AWS Machine Learning Specialty (Q3 2026).

TECHNICAL SKILLS

Languages: Python (10+ yrs) • SQL • Scala • JavaScript / Node.js

AI/ML & GenAI: LLMs • RAG • Prompt Engineering • LangChain • LangGraph • CrewAI • DSPy • TruLens • PyTorch • TensorFlow • scikit-learn • YOLO • OpenCV • XAI • NLP • Time-Series • Anomaly Detection • Active Learning • Voice AI (STT/TTS)

Cloud & MLOps: AWS (Bedrock • SageMaker • Lambda • API Gateway • Aurora • CloudWatch • EC2 • S3 • EMR • Glue • Athena • EKS • MSK • Kinesis • SQS/SNS) • Azure (OpenAI • AI Search • Document Intelligence • Blob Storage • Container Apps • DevOps) • Snowflake (Cortex • Snowpark • Model Registry • Feature Store • Container Services • Tasks • Streams) • Kubernetes • Docker • Terraform • GitHub Actions • GitLab • Jenkins

Interfaces & Data: Streamlit • FastAPI • Django • Next.js • PostgreSQL • MySQL • MongoDB • BigQuery • Pandas • Apache Spark • Apache Airflow • Kafka

PROFESSIONAL EXPERIENCE

Managing Consultant — AI & Data Science • LPDG Group

Apr 2023 – Present • Germany & India • Utilities • Manufacturing • E-mobility • Retail • B2B

Voice AI Agents for Customer Service — Retail / E-commerce • Germany

- Deployed end-to-end voice AI agents (STT → LLM → TTS) handling inbound customer service calls autonomously; integrated with order management + inventory via tool calls for full resolution without human escalation. ~65% calls auto-resolved • multi-FTE workload absorbed • ~40% average handle-time reduction.
- Stack: Python • CrewAI • VAPI • n8n • Twilio • Azure OpenAI • Azure STT/TTS • Streamlit.

Intelligent Invoice Entity Recognition Platform — Utilities • India — \$100K+/month savings

- Built FTP → Azure Document Intelligence → LLM → Snowflake → Streamlit pipeline processing multi-format PDFs (1–1,000 pages); confidence-gated human review prevented silent errors; service-address normalisation against ground-truth source.
- Stack: Python • Snowflake • Azure Document Intelligence • Azure OpenAI • Streamlit • GitHub Actions CI/CD.

Reusable MLOps Forecasting Framework — 5 Use Cases, 1 Codebase — Manufacturing & E-mobility • Germany

- One parameterised pipeline (pre-process → train → infer → post-process) serving 5 assembly-line + e-mobility forecasting use cases with zero code duplication; Snowflake Model Registry + Feature Store; GitHub Actions CI/CD across all 5 pipelines.
- Eliminated costly Snowflake → Databricks → Snowflake round-trip; resolved missing sensor time-series via behavioural clustering — enabling reliable forecasting where previously impossible.

7 Production RAG Chatbots — 1 Plug-and-Play Framework — Multi-client

- Shipped 7 production chatbots from one client-agnostic framework (new client = new config, not new codebase); multi-turn conversation via LangChain Retrieval Agents + Conversation Memory; FAISS + Azure AI Search for semantic retrieval; FastAPI

async; Kubernetes auto-scaling + Terraform-provisioned infra. Led cross-functional team of 4 engineers + 3 client SMEs across all 7 deployments.

Applied Data Scientist · Vivere GmbH

May 2022 – Feb 2023 · Hamburg, Germany · 9-month fixed-term contract

- Built LLM-powered product intelligence dashboard (Azure OpenAI + BigQuery + PostgreSQL) — non-technical product teams query market data conversationally; competitor clustering, organic-vs-inorganic spend attribution, event-driven Asana/email alerting via FastAPI/Pydantic.

Data Scientist (Research) · IBM, Universitätsklinikum Magdeburg

Oct 2020 – Mar 2022 · Magdeburg, Germany

- Active learning on BraTS2017 medical imaging: Query-By-Committee (KLD · JSD · Entropy) achieved 7% improvement in informative sample selection over random; precision fluctuation reduced 12% → 1% — cutting medical-image annotation cost.
- Explainable Deep Reinforcement Learning: Dropout DQN + saliency maps on MNIST/Space Invaders; 3–6% performance improvement using only 2.5–10% training data via MMD-Critic selection — demonstrating data efficiency over data volume.
- IRTEX — Explainable Image Retrieval: heatmaps + saliency maps at set-level and individual level; 80% majority user approval (85% global, 65% local explanations) validated via user studies.
- MSc thesis: Human Brain Simulation (VisMem cognitive model on Nengo) · IEEE paper: IRTEX — Image Retrieval with Textual Explanations (XAI), ieeexplore.ieee.org/document/9582666.

Software Engineer — Data & Backend · 3DQR GmbH

Apr 2019 – Sep 2020 · Magdeburg, Germany

- Automated MySQL schema migration framework supporting diverse configurations; 200+ end-to-end test suite (Mocha/Chai) covering all server-side paths — enabled confident ship cadence without manual regression.

Software Engineer — Data & Backend · Careers360 · ConveGenius Group

Jan 2016 – Oct 2018 · Delhi / Noida, India

- LMS delivery managing team of 3 (Django · Node.js · MySQL · AWS); built 24×7 real-time student-query chat system; secure Android REST APIs with server-side business logic; analytics dashboarding for student-engagement and platform-health metrics.

EDUCATION

MSc Digital Engineering · Otto-von-Guericke-Universität Magdeburg, Germany · Apr 2019 – Mar 2022

BTech Computer Science · The LNM Institute of Information Technology (LNMIIT), Jaipur, India · Jul 2012 – May 2016

CERTIFICATIONS

Anthropic — Introduction to Agent Skills · SnowPro Core Certification · SnowPro Specialty: Gen AI Certification · In progress: Anthropic — Building with the Claude API; AWS Certified Machine Learning – Specialty (Q3 2026)

SELECTED PERSONAL PROJECTS

MindMorph — Multi-Agent Adaptive Learning Platform · github.com/ATV-TECH-SOLUTIONS/MindMorph_Learning_Platform — LangGraph + CrewAI orchestrator routing learner queries across specialised agent pipelines · multi-model routing (GPT · Claude · Gemini) by task type/latency/cost · RAG (Pinecone) with Playwright/Firecrawl grounding · DSPy + TruLens prompt evaluation · Kafka analytics loop · deployed on Kubernetes (EKS) with Prometheus/Grafana/OpenTelemetry observability.

TinyFish Job Engine — LLM Job Relevance Scoring · github.com/tarunlnmiit/tinyfish_job_engine — 1,000+ jobs scored daily at 70%+ precision · multi-provider LLM fallback (60% API cost reduction) · zero-shot feature extraction across 90+ ATS sources · ~95% uptime via incremental state management.

AI Job Engine — Job Search Automation · github.com/tarunlnmiit/ai-job-engine — 12 portal scrapers (LinkedIn · Greenhouse · Lever · Indeed · Naukri · Wellfound · Instahyre · ArbeitNow · Hirist · Remotev · WeWorkRemotely · Hacker News) · Gemini Flash fit scoring · dynamic resume tailoring per JD · Playwright one-click Easy Apply · Streamlit dashboard.

AI Content Machine — Multi-Platform Content Automation · github.com/tarunlnmiit/ai-content-machine — 45+ Python scripts publishing 200+ pieces/month across 7 platforms (YouTube · LinkedIn · Medium · Instagram · Twitter · Substack · podcasts) · 603-node knowledge graph preventing repeat coverage · 85% reduction in manual repurposing · Claude API + FFmpeg + n8n.

ADDITIONAL

Content & Community: 50+ technical articles on Medium · YouTube channel Breath of Data Science — practitioner guides.

Languages: English C1 · German A2 · Hindi Native